

PROJECT CONCEPT NOTE

MedFlow AI: Hospital Triage & Scheduling Portal

Document Version:	1.0
Author:	Development & AI Operations Team
Target Audience:	Stakeholders, Medical Administrators, Technical Operations

1. Executive Summary

In busy healthcare environments, immediate and accurate patient triage is a major challenge. MedFlow AI addresses this by decoupling patient routing and appointment scheduling from manual operations.

Using an Agentic AI workflow (built with LangGraph) and a fast API gateway (FastAPI), MedFlow AI processes patient symptoms, classifies them into correct care departments (Emergency, Mental Health, General Ward), dynamically books the next available time slot for matching on-call specialists, and presents it in a premium, glassmorphic client interface.

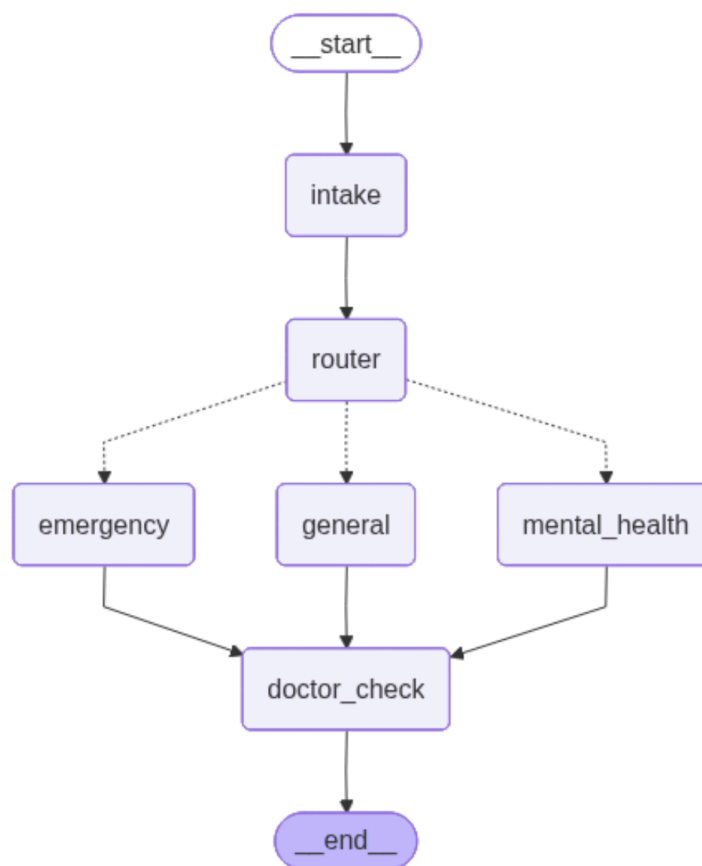
2. Problem Statement

- Hospital emergency rooms and clinics suffer from significant inefficiencies during the patient intake phase:
1. Bottlenecks in Manual Triage: Understaffed intake desks cause delays in evaluating patients, which can worsen critical health conditions.
 2. Human Routing Errors: Directing psychiatric crises or acute cardiac conditions to general clinics can lead to poor patient outcomes.
 3. Disjointed Doctor Scheduling: Medical staff coordinates schedules manually or via separate databases, making it difficult to allocate appointments dynamically.
 4. Outdated User Interfaces: Healthcare systems often rely on legacy software, which slows down intake coordinators and increases cognitive load.

3. Proposed Solution

- MedFlow AI is a modern clinical assistant platform split into two decoupled layers:
- * The Brain (AI Backend): A LangGraph state machine that evaluates patient intake details. It uses safety keyword lists (crisis and emergency bypass rules) and LLMs (Llama 3.3 via Groq) to route patients and calculate appointments.
 - * The Portal (Responsive Frontend): A modern, glassmorphic dashboard built using Vanilla CSS and JavaScript, allowing coordinators to quickly run evaluations, check doctors' schedules, and reset staff logs.

Agentic State Machine Diagram:



4. Core System Workflow

1. Patient Intake: The coordinator inputs the patient's name, age, and symptoms in the web app.
2. Safety Routing Guardrails: If emergency indicators (e.g. chest pain) are present, the patient is routed directly to Emergency. If crisis indicators (e.g. self-harm) are present, the patient is routed to Mental Health. Standard queries are processed by Llama 3.3.
3. Smart Scheduling: The system queries doctors.csv, identifies the specialist in the assigned ward with the earliest slot, calculates their next appointment slot, and saves the appointment details to patients.csv.
4. Visual Portal Update: The result card displays the routed ward, the doctor's name, the assigned time slot, and the reasoning behind the classification.

5. Technology Stack

- * Core Agent Engine: langgraph, langchain-core, and langchain-groq (Llama 3.3 70B model).
- * API Gateway & Middleware: fastapi and uvicorn (cross-origin CORS enabled).
- * Data Management: Python pandas operating on local CSV structures (doctors.csv, patients.csv).
- * Web UI Layout: Vanilla HTML5, CSS3, JavaScript (responsive UI, CSS transitions, custom SVG loader, modal schedule viewer).

6. Key Value Proposition & Impact

1. Improved Patient Safety: Automatic keyword triggers ensure that emergency and mental health patients bypass LLM reasoning queues for immediate care.
2. Reduced Wait Times: Automated time slot calculations assign patients to available specialists without scheduling conflicts.
3. Streamlined Coordination: Doctors' schedules are logged centrally. Staff can review any doctor's daily patient list with a single click in the Doctor Directory.
4. Modern Design System: The premium dark slate dashboard with glowing status badges reduces visual fatigue for administrative staff.

7. Future Expansion Roadmap

- * EHR Integration: Connect patient records directly to standard HL7/FHIR hospital databases.
- * Notification System: Send SMS and email reminders with confirmed slot times.
- * Telehealth Integration: Include links to video consultation interfaces directly on the assigned result cards.